

Question	Answer	Marks
6(a)	energy transferred (to the component) per (unit) charge	B1
6(b)(i)	$Q = It$ $= 0.030 \times 4.0 \times 60$	C1
	$= 7.2 \text{ C}$	A1
6(b)(ii)	$I = V / R$ $I_1 = 8.0 / (430+240)$	C1
	$= 0.012 \text{ A}$	A1
6(b)(iii)	$I_2 = 0.030 - I_1$ $= 0.030 - 0.012$ $= 0.018 \text{ A}$	A1

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6(b)(iv)	$R = V / I_2$ $= (8.0 - (0.018 \times 210)) / 0.018$	C1
	$= 230 \, \Omega$	A1
	OR resistance of top branch $= 8.0 / 0.018$ $R = 8.0 / 0.018 - 210$	(C1)
	$= 230 \, \Omega$	(A1)
	OR total circuit resistance $= 8.0 / 0.030 = 267$ $1 / 267 = 1 / (210 + R) + 1 / (430 + 240)$ $1 / 267 - 1 / 670 = 1 / (210 + R)$ $210 + R = 443$	(C1)
	$R = 230 \, \Omega$	(A1)
6(c)	(When) the galvanometer reads 0 (A)	M1
	The ratio of the resistances in the top branch will equal the ratio of the resistances in the bottom branch (so the resistance of X can be determined)	A1
	OR The ratio of the left pair of resistances will equal the ratio of the right pair of resistances	(A1)

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